

POC in ACE Framework

PROs and CONs of using ACE

PROs

1. Structured workflow management - ACE provides a clear framework for modeling and automating business process using BPMN, ensuring efficiency and compliance.
2. Modular and Scalable - Its architecture allows business to scale application and workflows without a complete system overhaul.
3. Different teams access the same ACE context - ACE has its own DB to hold context data which allows different team to share the same ACE context when they are on different infra.

CONs

1. Maintenance a separate ACE context - Need to send general context data to ACE, which can be duplicated from CRR table data.
2. Maintenance separate repos and CICD pipeline - bundled repo, deployment repo, BPMN files, which introduces extra vulnerabilities, deployment pipeline, maintenance work, etc.
3. Learning curve - User must understand BPMN modeling and process automation concepts, which may require training.
4. Dependence on vendor/ACE support

Existing CRR approval flow

User clicks on Submit button/link, the assessment will be marked as "SUBMIT" in RISK_ASSESS_EVENT_BTCH_STA_TRK table, which then moves to "INPROG" when the spark jobs starts and when the spark batch ends, it will be changed to "COMPLETE" status. User may view the result of the completed sandbox, once user trigger the Implement operation, the completed assessment becomes as Pending Approval 1 status, then it would need 2 users' approval to pass thru Pending Approval 1 and Pending Approval 2 stage util merging to Production, users may also reject the assessment.

Customer Risk Rating Definition		
> Risk Assessments		
Assessment Name:	FRANCE-Assessment-SBX-01212025	Description: France Production Assessment, do not delete
		History Implement
Assigned Categories		
Name	Description	
ARFs and HROs	This Category contains data elements that are the result of Additional Risk Factor and High Risk Override rules that have triggered for the customer	
Customer	This Category contains data elements relevant to attributes of the customers demographics	
Geography_1	This Category contains data elements relevant to Geographical locations in which the customer is involved.	
Products and Services	This Category contains data elements relevant to Products and Services for which the customer is the owner.	
Transactions	This Category contains data elements relevant to Transactions in which the customer is involved.	

Use cases in modernized CRR

CASE 1: User A submits the request

1. Frontend: verify the user entitlement (CRR-Sandbox-Submit-Only), call /assessment/submit API, send the assessment id, ecn, email_id, comment, etc.
2. Backend: mark the assessment as PENDING APPROVAL with ecn. Send email to audience

Case 2.1: User B approves the request

1. Frontend: verify the user entitlement (CRR-Sandbox-Edit), call /assessment/{approve} API, send the assessment id, ecn, email_id, comment, etc.
2. Backend: mark the assessment as APPROVED, send the email

Case 2.2: User B rejects the request

1. Frontend: verify the user entitlement (CRR-Sandbox-Edit), call /assessment/{reject} API, send the assessment id, ecn, email_id, comment, etc.
2. Backend: mark the assessment as REJECTED, send the email

Case 2.3: User B requires action and revert to initial status

1. Frontend: verify the user entitlement (CRR-Sandbox- Edit), call /assessment/{rollback} API, send the assessment id, ecn, email_id, comment, etc.
2. Backend: mark the assessment as INITIAL. Send email to audience